Weighing and Filling Machine



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Weighing and Filling Machine

What it is used for

A weighing and filling machine measures a target quantity of product and dispenses it into containers or packages. Correct calibration, product setup, and filling control help maintain consistent pack weight and reduce material waste.

Main parts

- Product hopper or supply tank
- Weighing platform, load cell, or weighing bucket
- Filling nozzle or discharge gate
- Feeding mechanism such as auger, pump, or vibratory feeder
- Control panel and target-weight settings
- Container holder, conveyor, and safety guards

Before you start

1. Read the operating instructions and confirm the machine is suitable for the product and container type.
2. Inspect the hopper, weighing system, filling nozzle, conveyor, guards, and electrical connections for damage.
3. Clean and sanitize product-contact surfaces according to the required hygiene procedure.
4. Check the weighing system calibration using approved test weights or the specified calibration method.
5. Set the target weight, allowable tolerance, feed speed, and filling mode for the product.
6. Ensure containers are clean, correctly positioned, and compatible with the filling method.

How to use

1. Load the approved product into the hopper or supply tank without exceeding its rated capacity.
2. Switch on the machine and allow the weighing system to stabilize as instructed.
3. Run the required zero or tare function with the empty weighing platform or container in position.
4. Place or feed an empty container into the designated filling position.
5. Start the filling cycle using the approved control or automatic sequence.
6. Monitor the product feed and confirm the machine stops or slows at the target quantity.
7. Check the filled container's weight using the machine display or an approved verification scale.
8. Adjust settings only within authorized limits if repeated checks show a consistent deviation.
9. Remove the filled container and transfer it to the next approved packaging or inspection stage.
10. Stop the machine, isolate power, and clean product-contact parts after the production run.

⚠ Safety

- Keep hands away from moving conveyors, augers, pumps, and filling mechanisms.
- Never bypass safety guards or reach into the hopper while the machine is operating.
- Disconnect and isolate power before clearing blockages, changing parts, or cleaning moving components.
- Use suitable lifting methods when handling heavy product containers or supply bags.
- Follow food hygiene and allergen-control procedures when changing products.
- Do not operate the machine if the load cell, guards, controls, or emergency stop are faulty.

Care & storage

- Clean the hopper, nozzle, weighing bucket, and other product-contact parts after use.
- Inspect the load cell, feed mechanism, seals, and nozzle for wear or buildup.
- Verify weighing accuracy at the scheduled frequency and after maintenance or relocation.
- Record calibration checks, weight deviations, cleaning, and maintenance as required.

Common mistakes to avoid

- Skipping tare or zeroing can cause every package to have an incorrect weight.
- Changing settings without verification can create overfilling or underfilling.
- Allowing product buildup around the nozzle can affect flow and fill accuracy.
- Ignoring calibration drift can cause out-of-tolerance packages and material losses.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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